Lecture 10: Sequences

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what a sequence is and how to write one using correct notation.
- ☐ Know what is meant for a sequence to converge or diverge.
- ☐ Find the limit of a sequence using different rules and techniques.
- ☐ Use the Sandwich Theorem and Continuous Function Theorem to evaluate limits.
- ☐ Recognize if a sequence is increasing, decreasing, or bounded and apply the Monotonic Sequence Theorem whenever possible.

Sequences

A sequence is a list of numbers in given order.

- Finite sequence: $\{a_n\}_{n=1}^{4} = \{2, 4, 6, 8\}$.
- Infinite sequence: $\{a_n\}_{n=1}^{\infty} = \{1, 4, 9, 16, \ldots\}$.

terms; 1st term; 2nd terms; $a_1 = 2$, $a_2 = 4$, $a_3 = 6$, $a_4 = 8$; index.

We can also see a sequence as a function whose domain is the set of positive integers. In our course, unless specified otherwise, we assume all sequences are real-valued.

Sequence is a function $f : \mathbb{Z}^+ \to \mathbb{R}$.

not continuous

Eg: $\{a_n\}_{n=1}^{\infty} = \{1, 4, 9, 16, \ldots\} = \{1, 4, 9, 16, \ldots, n^2, \ldots\}$

1^2 2^2 3^2 4^2

n^{th} term.

Representing Sequences

Instead of writing $a(n)$ for the function value, we write a_n to denote the nth term of the sequence.

Instead of $a : \mathbb{N} \to \mathbb{R}$, we denote a sequence by $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$ or $\{a_1, a_2, a_3, \ldots\}$.

Sequences can be described by writing rules that specify their terms (the nth term of the sequence).

Eg:

1. $\{a_n\} = \{2, 4, 6, 8, \ldots\} = \{2n\}_{n=1}^{\infty} = \{2(n-1)\}_{n=1}^{\infty}$
2. $\{a_n\} = \{1, 4, 9, 16, \ldots\} = \{n^2\}_{n=1}^{\infty}$
3. $\{a_n\} = \{1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}, \ldots\} = \left\{\frac{(-1)^{n-1}}{n}\right\}_{n=1}^{\infty} = \left\{\frac{(-1)^{n+1}}{n}\right\}_{n=1}^{\infty}$
4. $\{a_n\} = \{2, 2, 2, 2, \ldots\} = \{2\}_{n=1}^{\infty}$ ← constant sequence

In general, the index does not necessarily start from $n=1$.

Convergence and Divergence

$\lim_{n\to\infty} a_n = L$

The sequence $\{a_n\}$ converges to the number L if for every positive number ϵ there corresponds an integer N such that

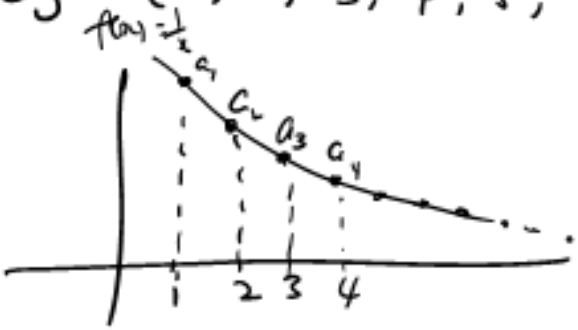
$$|a_n - L| < \epsilon \text{ whenever } n > N.$$

$\lim_{n\to\infty} a_n = \pm\infty$ or does not exists.

If no such number L exists, we say that $\{a_n\}$ diverges.

If $\{a_n\}$ converges to L, we write $\lim_{x\to\infty} a_n = L$ or $a_n \to L$, and we say that L is the limit of the sequence.

Eg: $\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots, \frac{1}{n}, \ldots\right\} = \left\{\frac{1}{n}\right\}_{n=1}^{\infty}$



$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{n} = 0$

or $\{a_n\} \to 0$

Compare with the def

$\lim_{x\to\infty} f(x) = L$

$\forall \varepsilon > 0, \exists M > 0$ such that

$|f(x) - L| < \varepsilon$ whenever

$x > M$

Example

Show that $\lim_{n\to\infty} \frac{1}{n} = 0$. ← $a_n = \frac{1}{n}$

Let $\varepsilon > 0$, we want to pick an N such that $N > \frac{1}{\varepsilon} > 0$

So for any $n > N$, we have

$$|a_n - 0| = \left|\frac{1}{n} - 0\right| = \frac{1}{n} < \frac{1}{N} < \varepsilon$$

Therefore, the sequence $\{a_n\} = \{\frac{1}{n}\}$ converges to 0.

↑ $\{\frac{1}{n}\}_{n=1}^{\infty}$

Example

Show that the sequence $\{\frac{n-1}{n+1}\}_{n=1}^{\infty}$ converges to 1.

(a_n, L)

Let $\varepsilon > 0$, we want to pick an N such that $N > \left(\frac{2}{\varepsilon} - 1\right) > 0$

Exercise.

(refer to last example)

So that for any $n > N$, we have

$$|a_n - L| = \left|\frac{n-1}{n+1} - 1\right| = \left|\frac{n-1}{n+1} - \frac{n+1}{n+1}\right|$$

$$= \left|\frac{n-1-n-1}{n+1}\right| = \left|\frac{-2}{n+1}\right|$$

$n+1$ +ve

Draft

$$= \frac{2}{n+1}$$

$n > N$

$$< \frac{2}{N+1} < \varepsilon$$

$$< \frac{2}{\left(\frac{2}{\varepsilon} - 1\right) + 1} = \varepsilon$$

$$\frac{2}{N+1} < \varepsilon$$

$$\Rightarrow \frac{N+1}{2} > \frac{1}{\varepsilon}$$

$$N+1 > \frac{2}{\varepsilon}$$

$$\boxed{N > \frac{2}{\varepsilon} - 1}$$

Example

Show that the sequence $\{1, -1, 1, -1, \ldots, (-1)^{n+1}, \ldots\}$ diverges.

Prove by contradiction.

Let $\{a_n\} = \{(-1)^{n+1}\}$

Suppose that $\{a_n\}$ converges to L.

$\forall \varepsilon > 0$, there must exists a number N such that for all $n > N$, we have $|a_n - L| < \varepsilon$.

Take $\varepsilon = 1/2$, $|a_n - L| < 1/2 \Rightarrow -1/2 < a_n - L < 1/2$

$\Rightarrow L - 1/2 < a_n < L + 1/2$

for all $n > N$

Consider when n is odd, $a_n = 1$

So $L + \frac{1}{2} > 1$

$\Rightarrow L > \frac{1}{2}$

Therefore $a_n \to L - \frac{1}{2} > 0$

Now, consider when n is even, $a_n \to -1$ (this contradicts with $a_n > 0$)

Therefore, $\{a_n\}$ must not be convergent, and hence $\sum$ is divergent.

Remark (grammar)

$\{a_n\}$ converges / diverges ← verb

$\sum a_n$ is convergent / divergent ← noun.

Diverges to (Negative) Infinity

In some cases, a sequence diverges because it get larger and larger. As n increases, a_n would become larger than any fixed number. We describe the behaviour of such sequence by the followings

The sequence $\{a_n\}$ diverges to infinity if for every number M, there is an integer N such that for all n larger than N, we have $a_n > M$. If this condition holds, then we write $n > N$

$$\lim_{n\to\infty} a_n = \infty \text{ or } a_n \to \infty.$$

Similarly, if for every number m there is an integer N such that for all n larger than N, we have $a_n < m$, then we say $\{a_n\}$ diverges to negative infinity and write

$$\lim_{n\to\infty} a_n = -\infty \text{ or } a_n \to -\infty.$$

diverge sequence $\lim_{n\to\infty} a_n = \infty$ or $-\infty$ Eg: $\{1,2,3,4,5,\ldots\} = \{n\} \to \infty$

diverge sequence $\lim_{n\to\infty} a_n$ does not exists Eg: $\{1,-1,1,-1,1,-1,\ldots\}$

Calculating Limits of Sequences

Suppose that $\{a_n\}$ and $\{b_n\}$ are convergent sequences with $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$. Then

- $\lim_{n\to\infty}(a_n + b_n) = A + B$ (converges) — sum rule.
- $\lim_{n\to\infty}(a_n - b_n) = A - B$ (converges) — difference rule
- $\lim_{n\to\infty}(k \cdot b_n) = k \cdot B$ (any number k) (converges) — constant multiple rule
- $\lim_{n\to\infty}(a_n \cdot b_n) = A \cdot B$ (converges) — product rule
- $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}$ if $B \neq 0$ (converges) — quotient rule

Calculus I

$\lim_{x\to\infty} f(x) = L$, $\lim_{x\to\infty} g(x) = M$

$\lim_{x\to\infty}(f(x) + g(x)) = L + M$

$\lim_{x\to\infty}(f(x) - g(x)) = L - M$

$\lim_{x\to\infty}(kf(x)) = kL$

$\lim_{x\to\infty}(f(x) \cdot g(x)) = L \cdot M$

$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \frac{L}{M}$ $M \neq 0$

Calculating Limits of Sequences

Examples:

(a) $\lim_{n\to\infty}(-\frac{1}{n}) = \lim_{n\to\infty}(-1)(\frac{1}{n}) = (-1)\lim_{n\to\infty}(\frac{1}{n}) = (-1)(0) = 0$

we cannot apply L'Hopital rule directly because $\{\frac{n-1}{n}\}$ is not a cts function.

(b) $\lim_{n\to\infty}(\frac{n-1}{n}) = \lim_{n\to\infty}(\frac{n}{n} - \frac{1}{n}) = \lim_{n\to\infty} 1 - \lim_{n\to\infty}\frac{1}{n} = 1 - 0 = 1$

(c) $\lim_{n\to\infty}\frac{5}{n^2} = \lim_{n\to\infty}\left(5\left(\frac{1}{n^2}\right)\right) = 5\lim_{n\to\infty}\left(\frac{1}{n^2}\right) = 5(0) = 0$

(d) $\lim_{n\to\infty}\frac{4-7n^6}{n^6+3} = \lim_{n\to\infty}\frac{4/n^6 - 7}{1+3/n^6} = \frac{0-7}{1+0} = -7$

(e) $\lim_{n\to\infty}(\sqrt{n^2+4n} - n) = \lim_{n\to\infty}(\sqrt{n^2+4n} - n)\,\frac{(\sqrt{n^2+4n}+n)}{(\sqrt{n^2+4n}+n)}$

$(a-b)(a+b) = a^2 - b^2$

$$= \lim_{n\to\infty} \frac{\left(\sqrt{n^2+4n}\right)^2 - n^2}{\sqrt{n^2+4n} + n}$$

$$= \lim_{n\to\infty} \frac{n^2+4n-n^2}{\sqrt{n^2+4n}+n} \qquad 4n \leftarrow n$$

$$= \lim_{n\to\infty} \frac{4n/n}{\frac{\sqrt{n^2+4n}}{n} + n/n}$$

$$\frac{n^2+4n}{n^2} = 1+\frac{4}{n}$$

$$= \lim_{n\to\infty} \frac{4}{\sqrt{\frac{n^2+4n}{n^2}}+1}$$

$$= \lim_{n\to\infty} \frac{4}{\sqrt{1+\frac{4}{n}}+1} \qquad \frac{4}{n} \to 0$$

$$= \frac{4}{\sqrt{1+0}+1} = 2$$

Calculating Limits of Sequences

Remark:

1. One consequence of the previous theorem is that if both $\{a_n\}$ and $\{b_n\}$ converges, their sum $\{a_n + b_n\}$ also converges. However, the converse is not true.

$\to A$ $\to B$ $\to A+B$

$\{a_n+b_n\}$ converges does not mean that $\{a_n\}$ and $\{b_n\}$ are both converges.

Eg: $\{a_n\} = \{1, 2, 3, \ldots\} = \{n\} \to \infty$ (diverges)

$\{b_n\} = \{-1, -2, -3, \ldots\} = \{-n\} \to -\infty$ (diverges)

$\{a_n+b_n\} = \{0, 0, 0, \ldots\}$

$= \{0\} \to 0$ (converges)

2. Another consequence is that every nonzero multiple of a divergent sequence $\{a_n\}$ diverges.

Eg: $\{a_n\} = \{1, 2, 3, \ldots\} = \{n\}$ diverges.

$\{2a_n\} = \{2, 4, 6, \ldots\} \to \infty$

Proof by contradiction

Suppose that there is a divergent sequence $\{a_n\}$ but $\{ca_n\}$ converges for some $c \neq 0$.

Let $k = \frac{1}{c} \neq 0$ $\{ca_n\}$ converges $\Rightarrow \{kca_n\}$ converges

$\Rightarrow \{\frac{1}{c} \cdot ca_n\}$ converges

$\Rightarrow \{a_n\}$ converges

(contradiction)

So, $\{ca_n\}$ diverges if $\{a_n\}$ diverges.

Calculating Limits of Sequences

The Sandwich Theorem for Sequences

Suppose that $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ are sequences such that $a_n \leq b_n \leq c_n$ holds for all n beyond some index N. If $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$.

Example:

- $\frac{\cos n}{n} \to 0$
- $\frac{1}{2^n} \to 0$
- $(-1)^n \frac{1}{n} \to 0$
- If $|a_n| \to 0$, then $a_n \to 0$
- $\frac{n!}{n^n} \to 0$

(This is valid because n is positive integer)

$-1 \leq \cos n \leq 1 \Rightarrow \frac{-1}{n} \leq \frac{\cos n}{n} \leq \frac{1}{n}$, $\to 0$, $\to 0$, $\to 0$

you can prove by induction.

$0 \leq \frac{1}{2^n} \leq \frac{1}{n}$, $\to 0$, $\to 0$, $\to 0$

$n=1$, $\frac{1}{2^1} = \frac{1}{2} < \frac{1}{1}$

$n=2$, $\frac{1}{2^2} = \frac{1}{4} < \frac{1}{2}$

$n=3$, $\frac{1}{2^3} = \frac{1}{8} < \frac{1}{3}$

2^n is grow faster than n

$$(-1)\frac{1}{n} \le (-1)^n \frac{1}{n} \le \frac{1}{n}$$

$\to 0$ $\qquad$ $\to 0$ $\qquad$ $\to 0$

$$(-1)^n = \begin{cases} 1, & n \text{ is even} \\ -1, & n \text{ is odd.} \end{cases}$$

$$-|a_n| \le a_n \le |a_n|$$

$\to 0$ $\qquad$ $\to 0$ $\qquad$ $\to 0$

$$0 \le \frac{n!}{n^n} = \frac{1 \times 2 \times 3 \times \cdots \times n}{\underbrace{n \times n \times \cdots \times n}_{n \text{ times}}} \le \frac{1}{n} \times \underbrace{\frac{n}{n} \times \frac{n}{n} \times \cdots \times \frac{n}{n}}_{n-1 \text{ times}}$$

$\frac{2}{n} \le \frac{n}{n}$, $\frac{3}{n} \le \frac{n}{n}$

$\to 0$ $\qquad$ $\to 0$ $\qquad$ $\to 0$ $\qquad$ $\to 0$ $\qquad$ $\to 1$ $\qquad$ $\to 1$ $\qquad$ $\to 1$ $\qquad$ $\to 0$

Calculating Limits of Sequences

The Continuous Function Theorem for Sequences

Suppose that $\lim_{n\to\infty} a_n = L$ and that f is continuous at L. Then

$\{a_n\} \to L$

$$\lim_{n\to\infty} f(a_n) = f(L)$$

$$\lim_{n\to\infty} f(a_n) = f\left(\lim_{n\to\infty} a_n\right)$$

Example: Find $\lim_{n\to\infty} \sin\left(\frac{\pi n^2}{4n^2+1}\right)$, if it exists.

We know that sine function is continuous everywhere on $\mathbb{R}$.

$$\lim_{n\to\infty} \sin\left(\frac{\pi n^2}{4n^2+1}\right) = \sin\left(\lim_{n\to\infty} \frac{\pi n^2}{4n^2+1}\right) = \sin\left(\lim_{n\to\infty} \frac{\pi}{4+\frac{1}{n^2}}\right)$$

$$= \sin \pi/4 = \frac{\sqrt{2}}{2}$$

Extra Example

① $\left\{\sqrt{\frac{n+1}{n}}\right\} \to 1$

Let $f(x) = \sqrt{x}$, f is cts on $[0, \infty)$

$$\lim_{n\to\infty} \sqrt{\frac{n+1}{n}} = \sqrt{\lim_{n\to\infty}\left(\frac{n+1}{n}\right)} = \sqrt{\lim_{n\to\infty}\left(\frac{1+\frac{1}{n}}{1}\right)} = \sqrt{1} = 1$$

② $\{2^{1/n}\} \to 1$

Let $f(x) = 2^x$, f is cts everywhere.

$$\lim_{n\to\infty} 2^{1/n} = 2^{\lim \frac{1}{n}} = 2^0 = 1$$

Using L'Hôpital's Rule

Theorem: Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence such that $a_n = f(n)$ for $n \geq n_0$. Then

If $\lim_{x\to\infty} f(x) = L$, then $\lim_{n\to\infty} a_n = L$

$$\lim_{n\to\infty} a_n = L \text{ whenever } \lim_{x\to\infty} f(x) = L.$$

limit of sequence (x cts. cannot apply L'Hôpital rule)

limit of function (we can use L'Hôpital rule if f is cts & differentiable.)

Note: This only works for finite limits.

$\lim_{x\to\infty} f(x)$ diverges does not mean that $\lim_{n\to\infty} a_n$ diverges.

Eg: $f(x) = x \sin(\pi x) \longrightarrow$ diverges.

$\{a_n\} = \{0\} \longrightarrow 0$ $\quad (\sin(n\pi) = 0)$

$\{a_n\} = f(n)$ for all positive integer n.

Example

Show that $\lim_{n\to\infty} \frac{\ln n}{n} = 0$.

Again, do not directly apply L'Hôpital Rule to the sequence.

sequence. ✗ cts. ✗ L'Hôpital Rule.

Let $f(x) = \frac{\ln x}{x}$, $x > 0$ $\quad a_n = f(n)$

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{\ln x}{x}$$

$$= \lim_{x\to\infty} \frac{1/x}{1} = 0$$

So, $\lim_{n\to\infty} \frac{\ln n}{n} = \lim_{x\to\infty} f(x) = 0$

Example

Suppose that $a_n = (1 + \frac{1}{n})^n$. Determine the limiting behaviour of a_n as $n \to \infty$.

$\lim_{n\to\infty} a_n$.

Let $f(x) = (1+\frac{1}{x})^x$ for $x > 0$

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} (1+\frac{1}{x})^x$$

$$= \lim_{x\to\infty} e^{\ln(1+\frac{1}{x})^x} = (1+\frac{1}{x})^x$$

Exponential function is the inverse of ln function.

$$= e^{\lim_{x\to\infty} \ln(1+\frac{1}{x})^x}$$

, exponential function is cts everywhere.

$$= e^{\lim_{x\to\infty} x \ln(1+\frac{1}{x})}$$

$$= e$$

$$= e^{\lim_{x\to\infty} \frac{\ln(1+\frac{1}{x})}{1/x}}$$

$$= e^{\lim_{x\to\infty} \frac{\frac{1}{1+1/x}\left(-\frac{1}{x^2}\right)}{\left(-\frac{1}{x^2}\right)}}$$, L'Hôpital rule.

$$= e^{\lim_{x\to\infty} \frac{1}{1+1/x}} = e^1$$

So $a_n \to e$ as $n \to \infty$.

Commonly Occurring Limits

1. $\lim_{n\to\infty} \frac{\ln n}{n} = 0$ (One of the previous examples)
2. $\lim_{n\to\infty} \sqrt[n]{n} = 1$ → Let $f(x) = \sqrt[x]{x}$, $x > 0$

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \sqrt[x]{x}$$

$x^{f(x)}$, $f(x)^{g(x)}$

$$= \lim_{x\to\infty} x^{1/x}$$

$$= \lim_{x\to\infty} e^{\ln x^{1/x}}$$

$$= e^{\lim_{x\to\infty} \frac{\ln x}{x}}$$

$$= e^{\lim_{x\to\infty} \frac{1/x}{1}} = e^0 = 1$$

3. $\lim_{n\to\infty} x^{1/n} = 1 \ (x > 0)$
4. $\lim_{n\to\infty} x^n = 0 \ (|x| < 1)$
5. $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$ (any x)
6. $\lim_{n\to\infty} \frac{x^n}{n!} = 0$

Eg: ① $\lim_{n\to\infty} \frac{\ln(n^2)}{n} = \lim_{n\to\infty} \frac{2\ln n}{n} = 2\lim_{n\to\infty} \frac{\ln n}{n} = 2(0)$

$= 0$

② $\lim_{n\to\infty} \sqrt[n]{n^2} = \lim_{n\to\infty} (n^2)^{1/n} = \lim_{n\to\infty} \left(n^{1/n}\right)^2 = \lim_{n\to\infty} \left(\sqrt[n]{n}\right)^2$

$= \left(\lim_{n\to\infty} \sqrt[n]{n}\right)^2$

Note that $f(x) = x^2$ is cts everywhere

$\lim_{n\to\infty} f(\sqrt[n]{n}) = f\left(\lim_{n\to\infty} \sqrt[n]{n}\right)$

$= (1)^2$

$= f(1) = 1^2 = 1$

$= 1$

③ $\lim_{n\to\infty} \sqrt[n]{3n} = \lim_{n\to\infty} (3n)^{1/n} = \lim_{n\to\infty} (3)^{1/n} (n)^{1/n}$

$= (1)(1) = 1$

④ $\lim_{n\to\infty} \left(-\frac{1}{2}\right)^n = 0$ $\quad |-\frac{1}{2}| = \frac{1}{2} < 1$

⑤ $\lim_{n\to\infty} \left(\frac{n-2}{n}\right)^n = \lim_{n\to\infty} \left(1+\frac{(-2)}{n}\right)^n = e^{-2}$

⑥ $\lim_{n\to\infty} \frac{100^n}{n!} = 0$

Bounded Monotonic Sequences

A sequence $\{a_n\}$ is bounded from above if there exists a number M such that $a_n \leq M$ for all n. Such number M is called an upper bound for $\{a_n\}$.

If M is the least among all upper bound of $\{a_n\}$, then M is the least upper bound.

A sequence $\{a_n\}$ is bounded from below if there exists a number M such that $a_n \geq M$ for all n. Such number M is called a lower bound for $\{a_n\}$.

If M is the greatest among all lower bound of $\{a_n\}$, then M is the greatest lower bound.

If $\{a_n\}$ is bounded from above and below, then $\{a_n\}$ is bounded.

Otherwise, we say $\{a_n\}$ is an unbounded sequence.

If $\{a_n\}$ is only bounded from above (or below) but not both, $\{a_n\}$ is unbounded.

Eg: $\{1, 2, 3, 4, 5, \ldots\}$ is bounded from below by 1 but it is not bounded from above, the sequence is unbounded.

Thm: A convergent sequence is bounded.

(If a sequence is convergent, then it is bounded.)

Proof: Let $\{a_n\}$ be a convergent sequence and $\lim_{n\to\infty} a_n = L$.

Then $\forall\, \varepsilon > 0$, there is a positive number N such that whenever $n > N$, $|a_n - L| < \varepsilon$.

Take $\varepsilon = 1$, $|a_n - L| < 1$

$L - 1 < a_n < L + 1$

which means, $L - 1 < a_{N+1} < L + 1$

$L - 1 < a_{N+2} < L + 1$

$\vdots$

$$L-1 < a_n < L+1 \quad \text{for } \forall\ n > N.$$

What about $a_1, \ldots, a_N$?

Obviously, $\min\{a_1, \ldots, a_N\} \leq a_i \leq \max\{a_1, \ldots, a_N\}$ for $1 \leq i \leq N$

So, $\{a_n\}$ is bounded from above by

$\max\{a_1, \ldots, a_N, L+1\}$ and

bounded from below by $\min\{a_1, \ldots, a_N, L-1\}$.

So $\{a_n\}$ is bounded.

Bounded Monotonic Sequences

A sequence $\{a_n\}$ is nondecreasing if $a_n \leq a_{n+1}$ for all n.

A sequence $\{a_n\}$ is nonincreasing if $a_n \geq a_{n+1}$ for all n.

The sequence $\{a_n\}$ is monotonic if it is either nondecreasing or nonincreasing.

Eg: $\{a_n\} = \{1, 2, 3, \ldots\}$ is nondecreasing. — monotonic

$\{a_n\} = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\}$ is nonincreasing. — monotonic

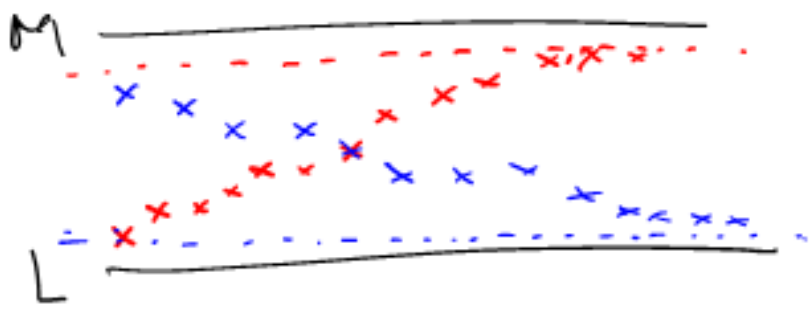
$\{a_n\} = \{0, 0, 0, \ldots\}$ is both nondecreasing and nonincreasing.

$\{a_n\} = \{1, -1, 1, -1, 1, -1, \ldots\}$ is neither nondecreasing nor nonincreasing. (non-monotonic)

Bounded Monotonic Sequences

The Monotonic Sequence Theorem

If a sequence $\{a_n\}$ is both bounded and monotonic, then the sequence converges.



$L \le \{a_n\} \le M$

$\{a_n\}$ is nondecreasing.

$\{a_n\}$ is nonincreasing.

However, a convergent sequence is not necessary monotonic.

Eg: $\{1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}, -\frac{1}{6}, \dots\} \longrightarrow 0$

but it is not monotonic.

Bounded Monotonic Sequences

Let the sequence $\{a_n\}$ be defined recursively by

(a_n is defined based on the preceding terms.)

$$a_0 = 1, \quad a_{n+1} = \frac{1}{2}(a_n + 3) \quad \text{for all } n \geq 0.$$

Evaluate $\lim_{n\to\infty} a_n$, giving reasons.

$$a_{n+1} = \frac{1}{2}(a_n + 3)$$

$$a_0 = 1$$

Base case:

$a_1 = \frac{1}{2}(1+3) = 2 > a_0$

$a_2 = \frac{1}{2}(2+3) = 5/2 > a_1$

$a_3 = \frac{1}{2}(5/2 + 3) = 11/4$

$\downarrow$ nondecreasing (we can prove this by induction)

Inductive hypothesis: Assume that $a_k \geq a_{k-1}$

Inductive step: $a_{k+1} = \frac{1}{2}(a_k + 3)$

$$= \frac{1}{2} a_k + \frac{3}{2}$$

$$\geq \frac{1}{2} a_{k-1} + \frac{3}{2}$$

$$= \frac{1}{2} (a_{k-1} + 3)$$

$$= a_k.$$

So by mathematical induction, $a_{n+1} \geq a_n$ and $\{a_n\}$ is nondecreasing (monotonic)

Obviously, $\{a_n\}$ is bounded from below by $a_0 = 1$

Claim: $a_n < 3$ for all n.

Base case: $a_0 = 1 < 3$, $a_1 = 2 < 3$

Inductive hypothesis: $a_k < 3$ $\quad a_k < 100$

Inductive step: $a_{k+1} = \frac{1}{2}(a_k + 3)$

$$= \frac{1}{2}a_k + \frac{3}{2}$$

$$< \frac{3}{2} + \frac{3}{2}$$

$$= 3$$

So by induction, $a_n < 3$ for all n.

So, $\{a_n\}$ is bounded from above by 3.

Therefore, A is bounded.

By Monotonic Sequence Theorem,

$\{a_n\}$ is convergent.

To find the $\lim_{n\to\infty} a_n$

Let $\lim_{n\to\infty} a_n = L$

$$\lim_{n\to\infty} \left(\frac{1}{2}(a_{n-1}+3)\right) = L$$

$$\frac{1}{2}\lim_{n\to\infty} a_{n-1} + \frac{3}{2} = L$$

$\lim_{n\to\infty} a_{n-1} \to L$

$$\frac{1}{2}L + \frac{3}{2} = L \Rightarrow L = 3.$$